Customer Shopping Behaviour Analyst

1.Project Overview

This project analyses customers shopping behaviour using transactional data from 3900 purchases across various product category the goal is to uncovered insight into spending pattern customer segments, product preferences and subscription behaviour to guide strategic business decision

2.Dataset Summary

-- Rows: 3,900
-- Columns: 18
-- Key Features:
- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (item Purchased, Category, Purchase Amount, Season, Size, Colour)

-- Shopping behaviour (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating Shipping Type)
- Missing Data: 37 values in Review Rating column

3.Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading**: Imported the dataset using pandas
- **Initial Exploration**: Used df.info() to check structure and .describe() for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900.000000	3900	3900
2	NaN	6	7
No	NaN	PayPal	Every 3 Months
2223	NaN	677	584
NaN	25.351538	NaN	NaN
NaN	14.447125	NaN	NaN
NaN	1.000000	NaN	NaN
NaN	13.000000	NaN	NaN
NaN	25.000000	NaN	NaN
NaN	38.000000	NaN	NaN
NaN	50.000000	NaN	NaN

- **Missing Data Handling**: Checked for null values and imputed missing values in the Review Rating columns using the median rating of each product category
- **Column Standardization**: Renamed columns to snake case for better readability and documentation
- **Feature Engineering**:
    - Created age_groups column by binning customer ages.
    - Created purchase_frequency_days column from purchase data
- **Data Consistency Check**: Verified if discount_applied and promo_code_used were redundant: dropped promo_code_used.
- **Database integration**: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis

4.Data Analysis using SQL

We performed structured analysis in PostgreSQL to answer Key business questions

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2.**High-Spending Discount Users** – Identified customers who used discounts but still spend above the average purchase amount

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

Total rows: 839 Query complete 00:00

3. **Top 5 Products by Rating** –Found products with the highest average review rating

	item_purchased text	Average product rating numeric
1	Gloves	3.8614
2	Sandals	3.8444
3	Boots	3.8188
4	Hat	3.8013
5	Skirt	3.7848

4.**Shipping Type Comparison** – Compared average purchase amounts between Standard and Express Shipping

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5**. Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6.**Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7.**Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchases history

	customer_segment text	Number of customer bigint
1	Loyal	2422
2	New	83
3	Returning	1395

8. **Top 3 Products per Category** – Listed the most perchased products with each category

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & subscriptions** -- Checked whether the customers with >5 purchases are most likely to subscribe

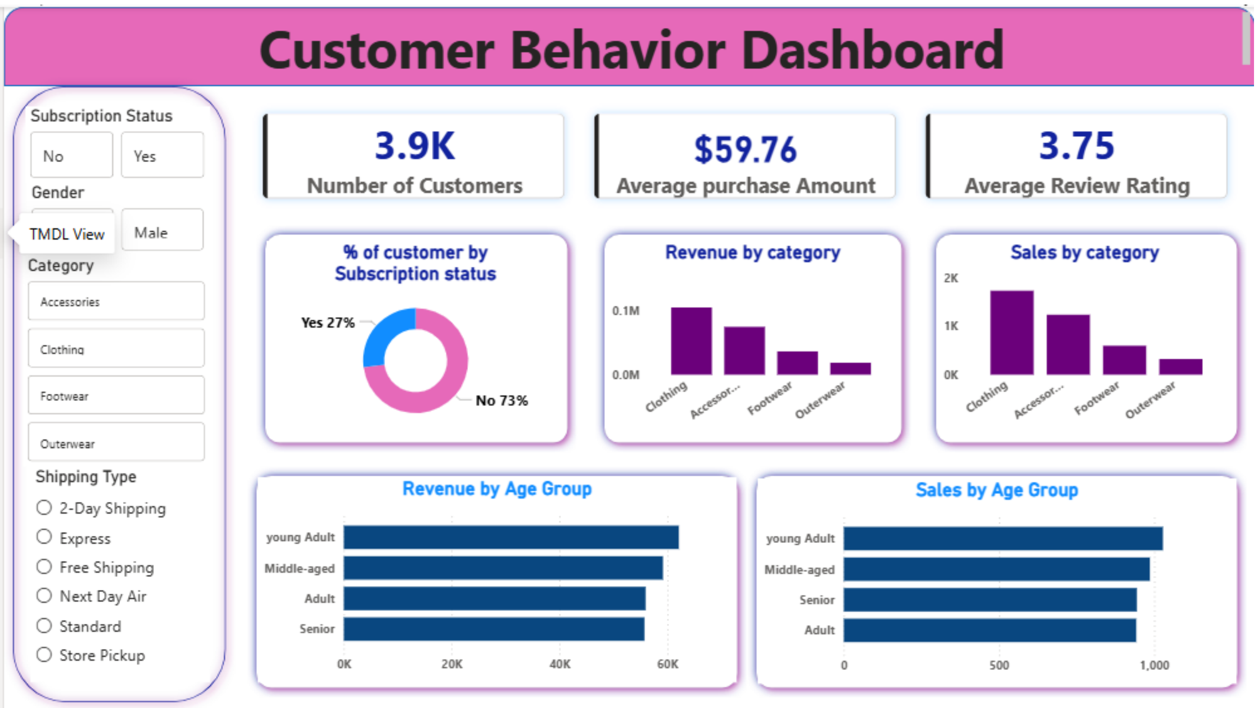
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group

	age_group text	total_revenue numeric
1	young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, we build an interactive dashboard in Power BI to present insights visually



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers
- **Customer Loyality Programs** – Reward repeat buyers to move them into the "Loyal" segment
- **Review Discount Policy** – Balance sales boosts with margin control
- **Product Positioning** – Highlight top – rated and best – selling products in campaign
- **Targeted Marketing** – Focus effort on high – revenue age group and express – shipping users